# Term 1 answer key

Formulas printed for students in every question. Answers with π are rounded to 1 decimal place, as each question says; probabilities are fractions in simplest form; every other answer is exact.

| WEEK | SESSION | Q1 A Angles & shapes | Q2 B Measurement | Q3 C Number & percentages | Q4 D Algebra, ratio & rates | Q5 E Number plane, data & chance |
|---|---|---|---|---|---|---|
| 1 | 1 | 27° | 50.3 m | −3 | $4m + 36$ | 164 |
| | 2 | 116° | 42 $cm^3$ | 6.4 | $300 and $200 | 5/8 |
| | 3 | 75° | 60 $m^2$ | 75% | $x = 5$ | −4, 2 |
| 2 | 1 | 50° | 52 cm | 16 | 15 | 13 |
| | 2 | 110° | 39 $m^2$ | 30 L | 60 km/h | 1/2 |
| | 3 | 37° | 4 km | 16/25 | $2 per kg | 9/25 |
| 3 | 1 | 143° | 50 $cm^2$ | 10 | $14 + 7k$ | 14 |
| | 2 | 115° | 36 $cm^2$ | 95% | 1 h 35 min | 12 |
| | 3 | 72° | 12.6 m | −12 | $x = 8$ | 3/5 |
| 4 | 1 | 65° | 8 cm | $35 | 16 | (3, 5) |
| | 2 | 70° | 34 m | $36 | 3 : 5 | 1/2 |
| | 3 | 127° | 5 m | 90% | 48 and 36 | 9 |
| 5 | 1 | 110° | 7 m | 30.6 | $15x − 10$ | 21/50 |
| | 2 | 123° | 75 $cm^2$ | 26% | 240 km | 7 |
| | 3 | 115° | 12 m | $56 | Bag A ($3 per kg) | 2 |
| 6 | 1 | 49° | 12 $m^2$ | 43 kg | $x = 16$ | 3/10 |
| | 2 | 115° | 44.0 cm | 15 | 21 | 7/10 |
| | 3 | 100° | 48 m | 17 | 00:30 | (4, 5) |
| 7 | 1 | 42° | 400 $m^2$ | 0.4 | $5x + 10$ | 19 cm |
| | 2 | 105° | 33 $cm^2$ | 25% | 2 : 3 | 15 |
| | 3 | 112° | 30 $cm^2$ | 18 kg | 18 and 30 | 7 |
| 8 | 1 | 82° | 72 $cm^3$ | 70 m | $y = 23$ | 9/25 |
| | 2 | 65° | 42 $cm^2$ | 250 | 14 | 7/9 |
| | 3 | 48° | 31.4 cm | 0.64 | 2 m/s | 2/11 |
| 9 | 1 | 56° | 2500 mL | −13 | $2x + 14$ | 6, 2 |
| | 2 | 125° | 30 cm | 20% | 12 km per litre | 2 |
| | 3 | 48° | 196 $cm^2$ | 18 | 35 minutes | 9 |
| 10 | 1 | 66° | 30 $cm^2$ | 41 | $y = 6$ | 8 |
| | 2 | 55° | 84 $m^2$ | $36 | 14 | 1/4 |
| | 3 | 140° | 80 $cm^2$ | 3/5 | 3 : 4 | 3/8 |

**A** MA4-ANG-C-01 complementary and supplementary angles, angles at a point, vertically opposite angles, angles on parallel lines; MA4-GEO-C-01 angle sum of a triangle and a quadrilateral, exterior angle of a triangle

**B** MA4-LEN-C-01 perimeter, circumference, length units; MA4-ARE-C-01 area of rectangles, squares, triangles, parallelograms and composite shapes, area units; MA4-VOL-C-01 volume of rectangular prisms, volume and capacity units

**C** MA4-INT-C-01 order of operations with integers; MA4-IND-C-01 powers, square and cube roots; MA4-FRC-C-01 fraction and percentage of a quantity, one quantity as a percentage of another, fractions, decimals and percentages, multiplying and dividing by 10, 100 and 1000*

**D** MA4-ALG-C-01 substitution, the distributive law; MA4-EQU-C-01 one- and two-step equations; MA4-RAT-C-01 simplifying ratios, sharing in a ratio, rates, best buys, speed; 24-hour time and time differences*

**E** MA4-LIN-C-01 tables of values, reading points on the number plane; MA4-DAT-C-01 dot plots; MA4-DAT-C-02 mean, median, mode and range; MA4-PRO-C-01 probability, complementary events, experimental probability*

Codes are from the NSW Mathematics K–10 Syllabus (2022). The NESA website could not be reached from the build machine, so **none of these codes was confirmed on the NESA site**; they match the codes used in the Year 9 and Year 10 keys.

* the topic is in this outcome as far as could be checked, but where it sits in the syllabus (or, for 24-hour time, whether it is Stage 3 revision with no Stage 4 code) could not be confirmed.